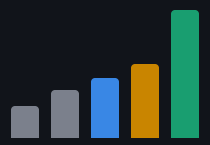
HIT SONG PREDICTION · TECHNION · PROJECT IN AI

What really predicts a hit?

Artist history, era context and shortcut features — tested on four datasets, with every gain checked for chance and for artefacts.

Lior Shalon & Nathanael Uzzan

Technion – Israel Institute of Technology · Project in Artificial Intelligence



ROADMAP

The project in six steps

1

Problem & data

Why published accuracies cannot be compared; the four datasets we use.

2

Method

The starting features, the two signals we add, the algorithms, and how every gain is tested.

3

Results on the benchmark

+5.0 points with every classifier, every decade, and against the only prior model.

4

Is it real?

How the benchmark was built, independent datasets, unseen and new artists.

5

What we tried and removed

Titles, rich audio, FFT, genre, lyrics: what we keep, what we drop, and why.

6

Time series

Components and forecasting of the Hot 100, and predicting big hits from their first weeks.

THE PROBLEM

Hit-prediction papers report 86 %, 88 %, 89 % — and none of these numbers can be compared

≠

Different datasets

Each study builds its own hits and non-hits. An 88 % on one set says nothing about another.

1×

One split, no test

Gains are measured on a single train/test split, usually without a significance test.



Shortcuts

A feature can separate hits from non-hits because of **how the negatives were chosen**, not because of anything that makes a song succeed.

Our question: which gains are **real** — and how big are they once all three problems are controlled?

DATA

Four datasets — the definition of "hit" changes every time

DATASET	SONGS	HITS	HIT LABEL	HOW NON-HITS WERE CHOSEN
Spotify Hit Predictor (Kaggle, 1960–2019)	40,560	49 %	Weekly Hot 100 in that decade	Rules: artist *and* genre absent from the hit list
audio subset (30 s Deezer clips)	3,562	52 %	same	same
ISMIR 2019 (Zangerle et al., peer-reviewed)	11,662	50 %	Hot 100 match	Random Million Song Dataset tracks
SpotGenTrack	101,909	19 %	Spotify popularity ≥ 55	Everything else
~~Billboard year-end 2000–2023~~	3,397		DROPPED 2,911 rows have no audio features	

Paper · Table I

METHOD · THE STARTING POINT

We start from the 13 audio descriptors that Spotify computes for every song

`danceability` `energy` `key` `loudness` `mode` `speechiness`
`acousticness` `instrumentalness` `liveness` `valence` `tempo`
`duration` `time signature`

Each song is a row of 13 numbers; the label is **1 = hit** (on the weekly Hot 100 in its decade), **0 = non-hit**. The task is **binary classification**.

The dataset also has *chorus_hit* and *sections*; we test them in Part 5 and they add nothing.

BASELINE · RANDOM FOREST ON THE 13 DESCRIPTORS

78.2%

accuracy, 5×5 cross-validation, 40,560 songs (49 % hits). Every gain in this talk is measured **against this number**.

Two cheap, non-audio signals added to the 13 Spotify audio features

ERA

Decade / release year

A hit in the 1960s does not sound like a hit in the 2010s. The era lets the model apply **era-specific rules**.

One-hot decade on the Hit Predictor; year or decade as a number elsewhere.

ARTIST HISTORY

Earlier Hot 100 songs

```
h(s) = log(1 + #hits by the same artist
strictly before the song's period)
```

Strictly earlier periods only → **no information from the song's own period or the future.**

Artist names normalised (accents, "feat.", "&" removed). 99.99 % of hits are found in the chart history.

One main model, four others to check that the result is not tied to it

ALGORITHM	WHY WE USE IT	AUDIO	+ DECADE + ARTIST	VERDICT
Random Forest 80 trees, 8 features per split	Configuration of the reference paper (Middlebrook & Sheik), so results compare; learns non-linear rules (decade × audio). **Never tuned**: a gain must come from the features.	78.0	83.2	KEEP main model for every experiment
XGBoost	The strongest learner for tabular data.	78.8	84.5	Best average, but ties the Random Forest on the canonical split (p = .87). Reported, not needed.
SVM (RBF kernel)	A kernel method, very different from trees.	77.3	82.8	Same gain; slower, no probabilities.
k-NN (k = 25)	Distance-based: “a song is a hit if similar songs were”.	75.0	79.6	Same gain; weaker overall.
Logistic regression	Linear and interpretable; also one of Du’s models.	72.7	78.2	Weakest: the relations are not linear. Kept for the head-to-head.

All five gain **+4.6 to +5.7 points**: the improvement comes from the features, not the model. External datasets use a 200-tree Random Forest. 5-fold CV · Table IV

Feature extraction, explanation and statistics

TURNING AUDIO INTO FEATURES

librosa: 185 descriptors from 30-second clips (timbre, rhythm, harmony).

FFT: 31 spectral descriptors (energy per frequency band).

PCA: compresses the 185 librosa descriptors to 40 components.

Essentia: the 592 descriptors of the ISMIR dataset.

TEXT AND EXPLANATION

TF-IDF + logistic regression: turns the song title into one probability, trained out-of-fold so the test songs are never seen.

TreeSHAP: shows which features the Random Forest relies on: artist history #1, decade #2, instrumentalness #3.

TESTING EVERY GAIN

McNemar: compares two models song by song on one split.

Corrected t-test (Nadeau–Bengio): compares two models over 25 folds without over-confidence.

Bonferroni: ~60 tests, so a result needs $p < 8{\cdot}10^{-4}$ to count as solid.

METHOD · HOW EVERY GAIN IS TESTED

Two estimates for every comparison — and a test that is not fooled by overlap

1 · Canonical split

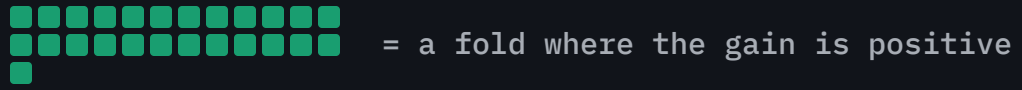
The split of our earlier experiments (seed 42; 20,000 train / 2,000 test). Two models compared song by song with **McNemar's test**.



Reproduces the old numbers exactly: .773 → .822.

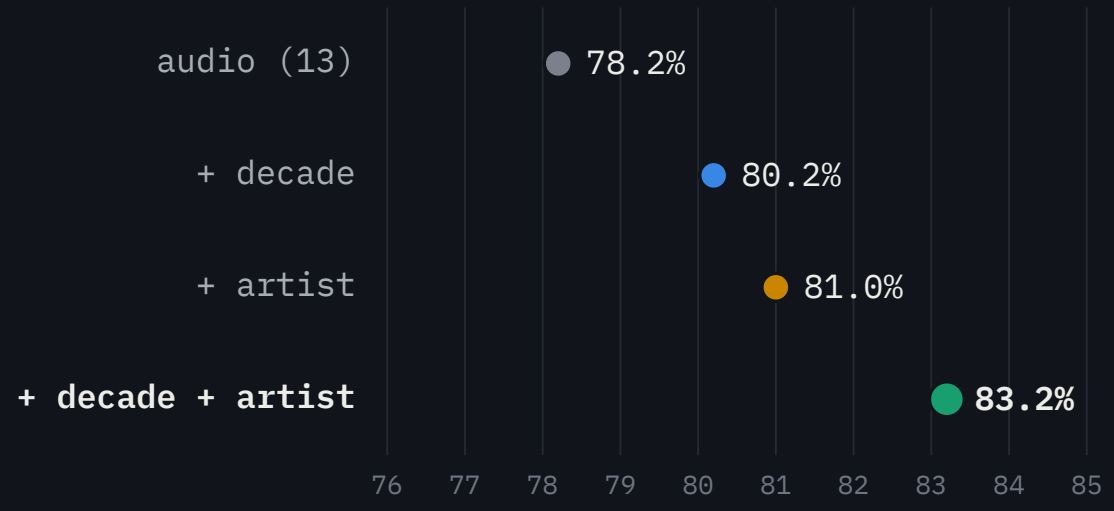
2 · Repeated cross-validation

5 folds × 5 repeats = **25 test sets**. Mean, 95 % confidence interval, and the **Nadeau–Bengio corrected t-test** — the naive t-test is over-confident because the folds overlap.



Scaling, PCA and title models are fitted inside each training fold only.

Decade and artist add +5.0 points, and the gains stack



Random Forest · 5×5 CV mean accuracy · paper Table II

83.2%

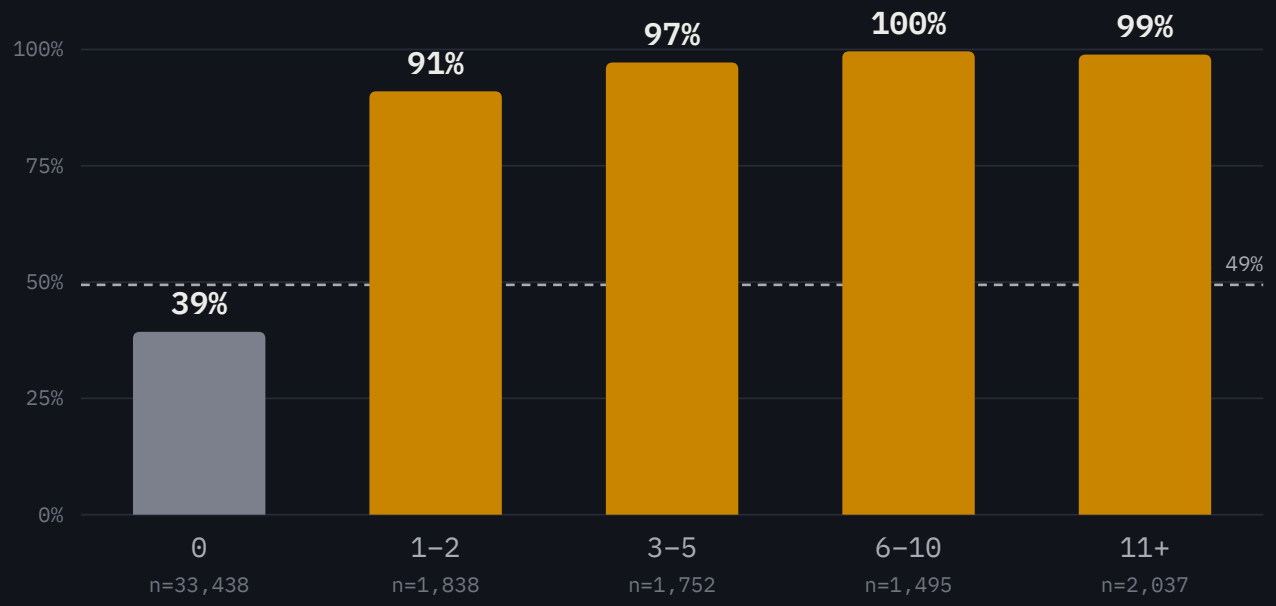
from 78.2 % with audio alone · ROC-AUC .859 → .918

25 / 25 folds
$p = 7 \cdot 10^{-17}$

F1 of the hit class .787 → .831 (precision .758 → .828, recall .819 → .833).

Canonical split: .773 → .796 → .800 → .822. The two signals alone reach only 67.0 %: they complement the audio, they do not replace it.

One earlier hit changes everything: 39 % → 91 %



Share of songs that are hits, by the artist's Hot 100 songs in earlier decades · paper Fig. 1

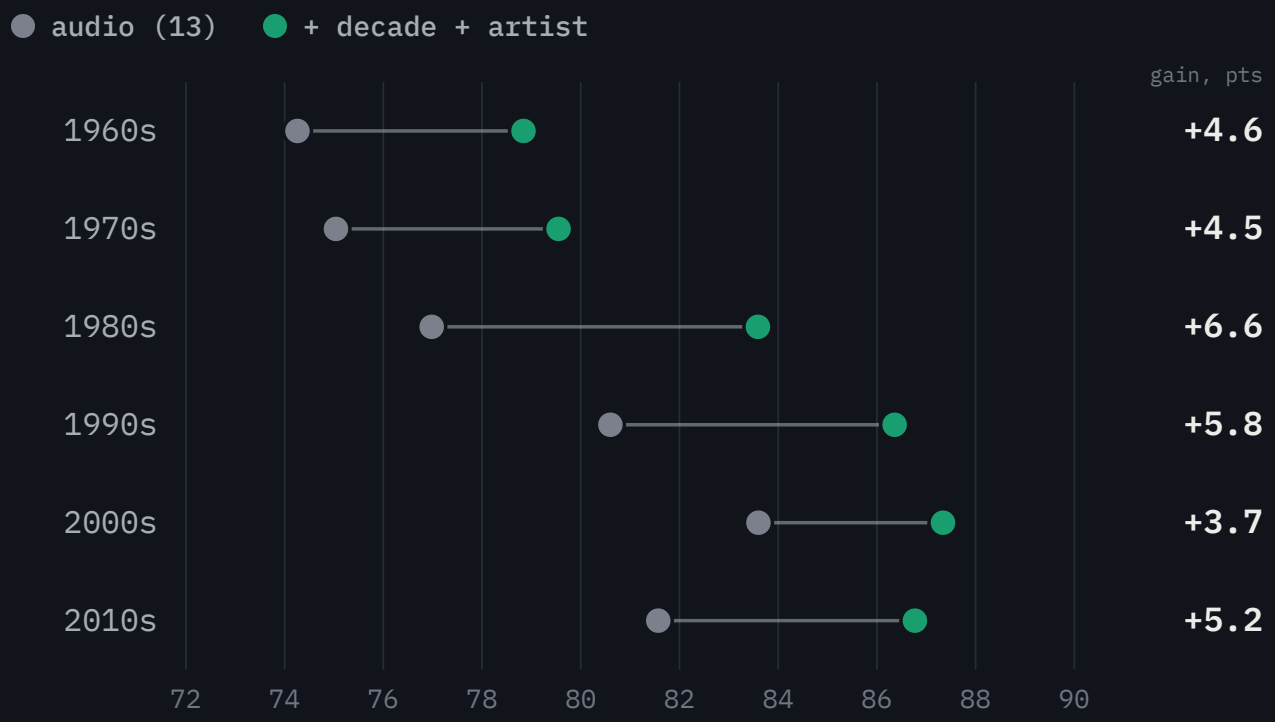
A strong, very non-linear relation: the first hits matter most, then it plateaus near 100 %.

MEAN EARLIER HITS

hits 3.21 · non-hits 0.037

Keep that 0.037 in mind — it comes back in Part 4.

The gain holds in all six decades, from +3.7 to +6.6 points

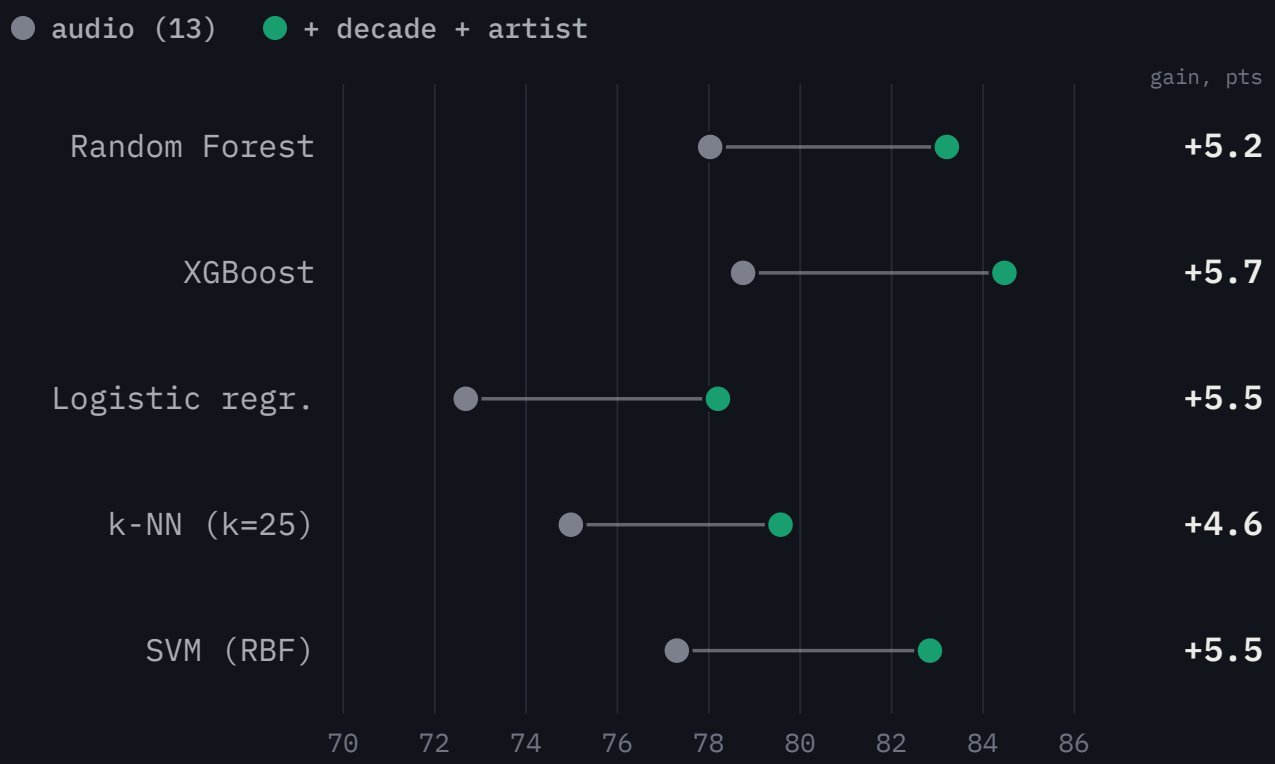


5×5 CV accuracy per decade · paper Table III

Each decade file is ~50 % hits, so the decade alone says nothing.

Its gain can only come from **interactions with the audio** — the model learns what a hit sounded like in that era.

Five very different models, the same +5 points: it is the features, not the model



5-fold CV, corrected t-test, all 5 folds improve · paper Table IV

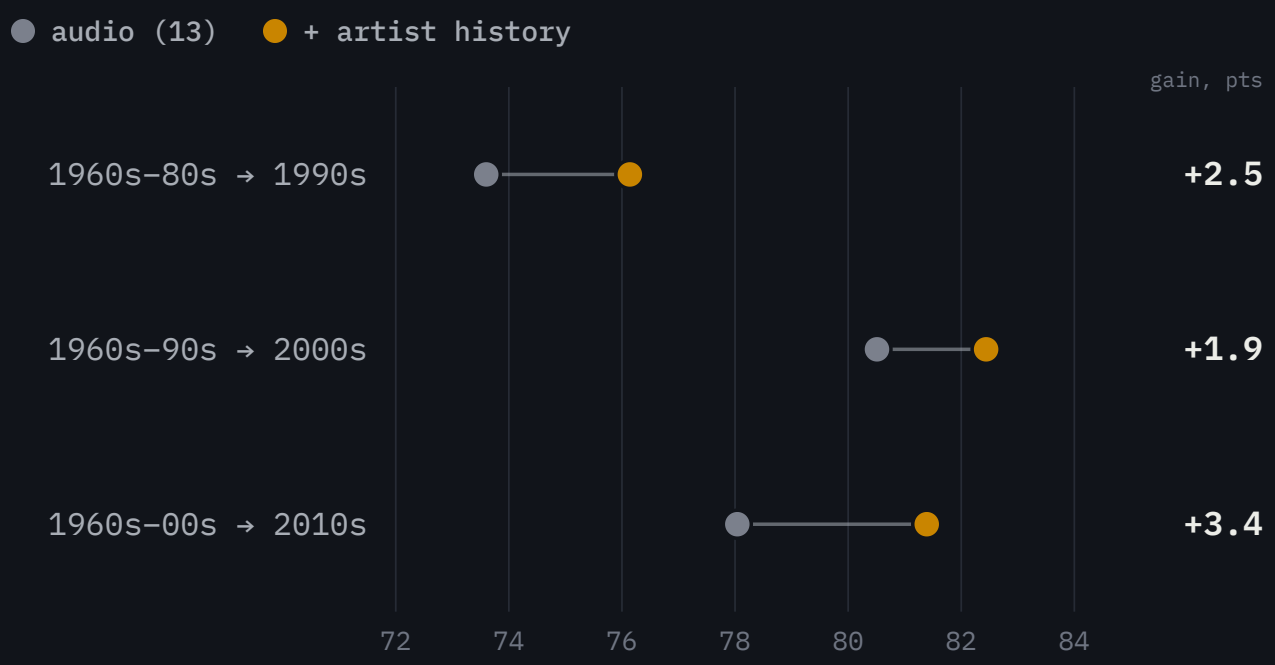
84.5%

XGBoost is the strongest on average — but on the canonical split it ties with the Random Forest (.820 vs .822, p = .87).

We keep the **Random Forest** as the main model: comparable with prior work, untuned, and within noise of XGBoost.

TreeSHAP ranks **artist history #1**, the six decade indicators #2 and instrumentalness #3.

Train on the past, predict a decade the model has never seen

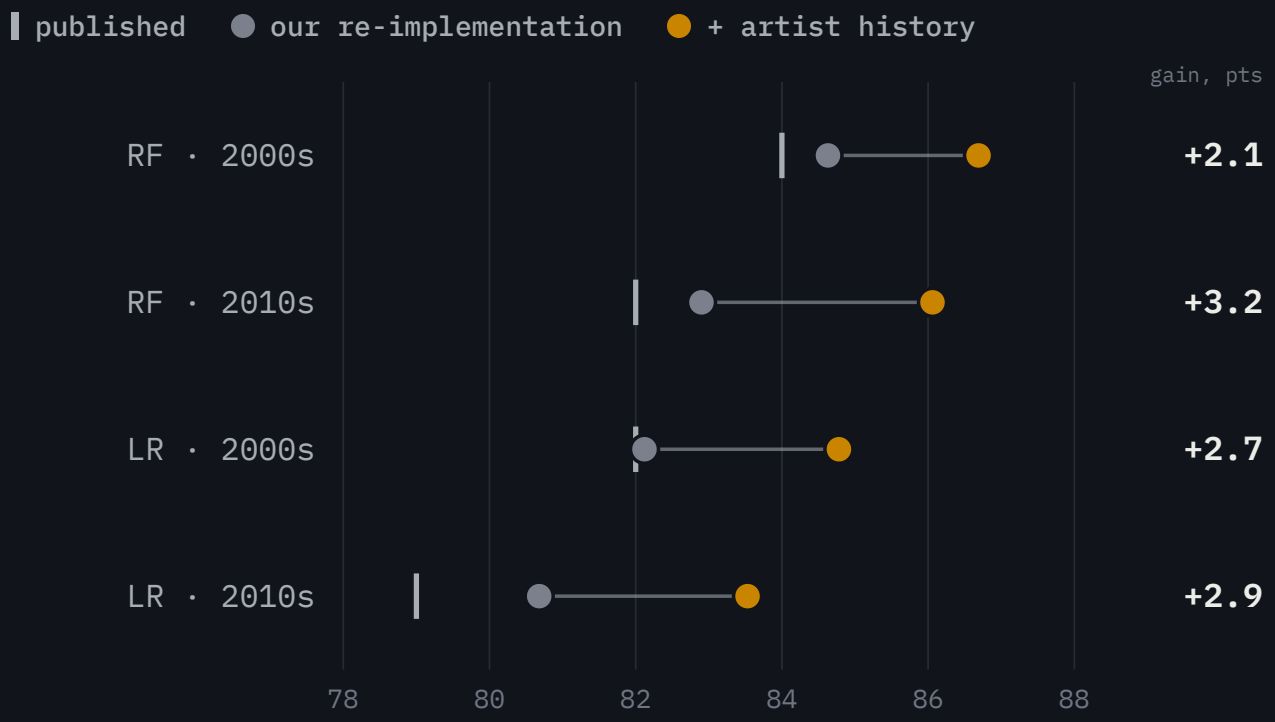


Random Forest, trained on all earlier decades · McNemar p between 10^{-7} and 10^{-21}

Artist history still adds **+1.9 to +3.4 points** on unseen decades, with similar gains for XGBoost.

The decade cannot be used here: the test decade never appears in training.

Their exact models and inputs, plus our feature: better on 50 out of 50 splits



Mean over 50 random 70/30 splits · paper Table V, Fig. 2

One model per decade, 13 audio + chorus_hit + sections — exactly as in the thesis. Our re-implementation lands **on or just above** their published numbers, which validates the protocol.

Adding artist history: **+2.1 to +3.2 points** on every split; the improved models beat the published accuracy on **100 %** of splits.

The Kaggle dataset chose its non-hits partly by artist

DATASET README — A “FLOP” MUST SATISFY ALL OF

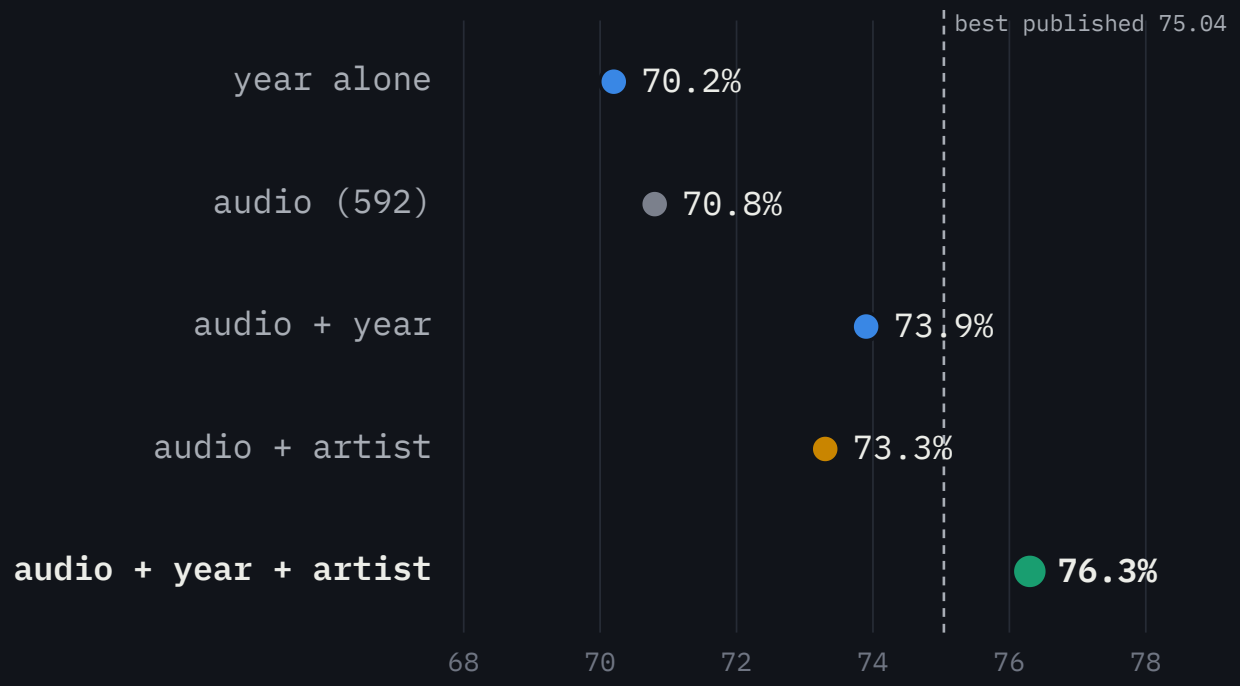
1. The track is not in that decade’s hit list.
2. **The track’s artist is not in that decade’s hit list.**
3. It belongs to a “non-mainstream and/or avant-garde” genre.
4. That genre has no song in the hit list.

96.6%

of songs by an artist with an earlier hit are hits. On this benchmark, “known artist” almost means “hit”.

So the **+5.0** is a **benchmark-specific upper bound**. We need data whose negatives are not chosen by artist → ISMIR 2019.

With artist history we reach 76.3 %, above the best published 75.04 %

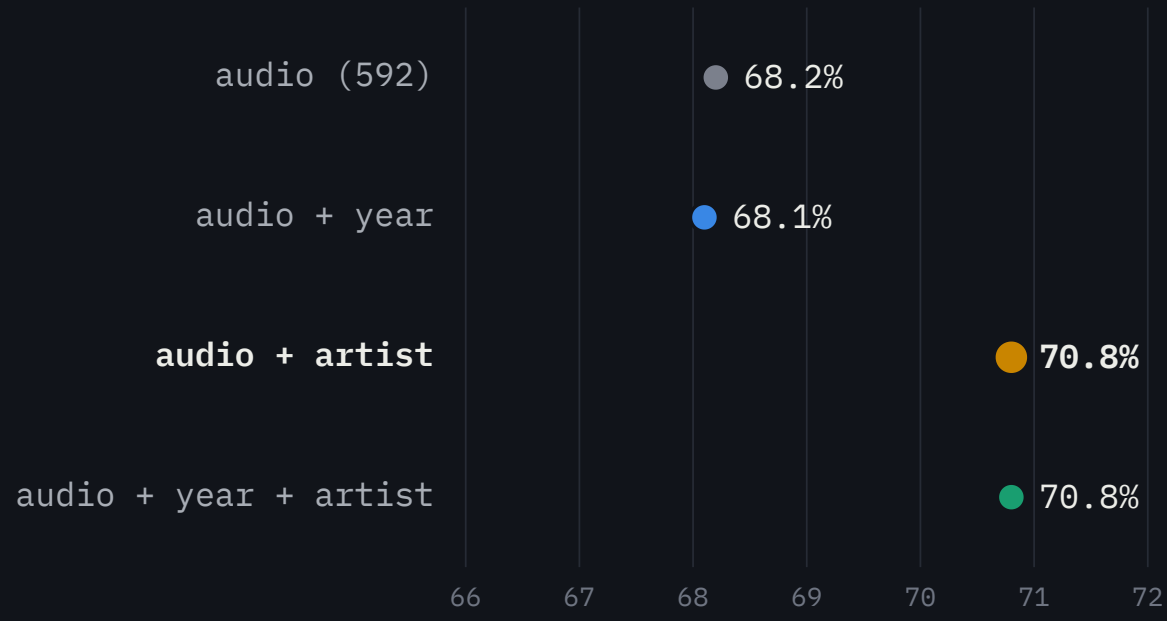


Random Forest on 592 Essentia descriptors · 5×2 CV mean · paper Table VI

Audio + year mirrors the inputs of the best published model and gives 73.9 %. Artist history adds **+2.4 points** (10/10 folds, $p = 7{\cdot}10^{-8}$).

But look at **year alone: 70.2 %** — as good as 592 audio descriptors. Hits in this dataset are older (median 1987) than non-hits (2001).

Match the years: release year becomes useless, artist history still adds +2.6



Same data, one non-hit from the same year for every hit · 5×2 CV · paper Table VI

+2.6

points, $p = 3{\cdot}10^{-5}$, 10/10 folds. Negatives chosen **neither by artist nor by date**.

This is the number we defend as the transferable value of artist history. Year: p = .66; year alone falls to 44.8 %, below chance.

EXTERNAL VALIDATION · SPOTGENTRACK, 101,909 TRACKS

With 81 % non-hits, accuracy hides the signal — ROC-AUC shows it



Artist history: AUC **.690 → .762** (+ decade .777). The 207 low-level audio features add nothing on top of it. Hits found: 8 % → 25 % (F1 .147 → .358). 5-fold CV · paper Table VII

ROBUSTNESS · FOLDS GROUPED BY ARTIST

Does the model just recognise artists? No — the gains survive when test artists are never seen

HIT PREDICTOR · decade + artist

+5.0 → +5.0

82.2 % grouped (83.2 % random). 25/25 folds, $p = 10^{-13}$.

ISMIR B · artist history

+2.6 → +2.9

10/10 folds, $p = 9{\cdot}10^{-7}$. Year still useless.

ISMIR A · audio + year + artist

75.9%

CI [75.2, 76.6] — still entirely above the published 75.04 %.

All accuracies drop by up to one point — the audio carried a little artist identity — but the gains remain (ISMIR A loses at most half a point). Paper Table VIII

ROBUSTNESS · COLD START

For a new artist, it is the era that helps — not the artist feature

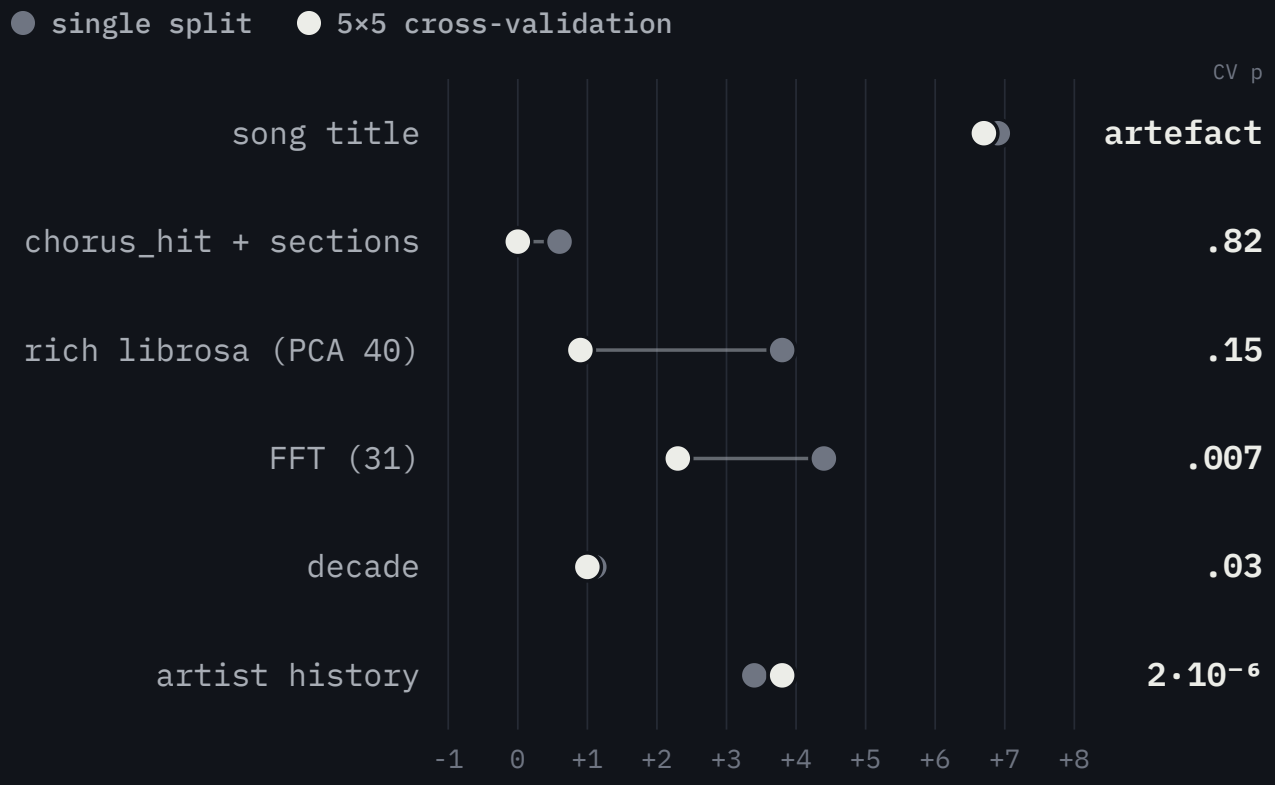
DATASET	HITS NEW / KNOWN	NEW · + ERA	NEW · + ARTIST	KNOWN · + ARTIST
Hit Predictor	39 % / 97 %	+2.2 $p=2\cdot10^{-9}$	+0.4 n.s.	+15.0 $p=3\cdot10^{-14}$
ISMIR A	36 % / 75 %	+3.4 $p=6\cdot10^{-5}$	+0.8 n.s.	+4.4 $p=.003$
ISMIR B (year-matched)	37 % / 72 %	+0.1 n.s.	+1.4 $p=.002$	+5.3 $p=8\cdot10^{-5}$

Gain in points over audio, artist-grouped CV. On the benchmark, artist history's whole gain comes from the 17.6 % of songs by established artists.

Paper Table VIII

AUDIT · WHICH GAINS ARE NOT REAL

A big, significant gain can still be wrong: artefact or luck



Gain over audio in points · top: full benchmark · bottom: 3,562-song audio subset · paper Table IX

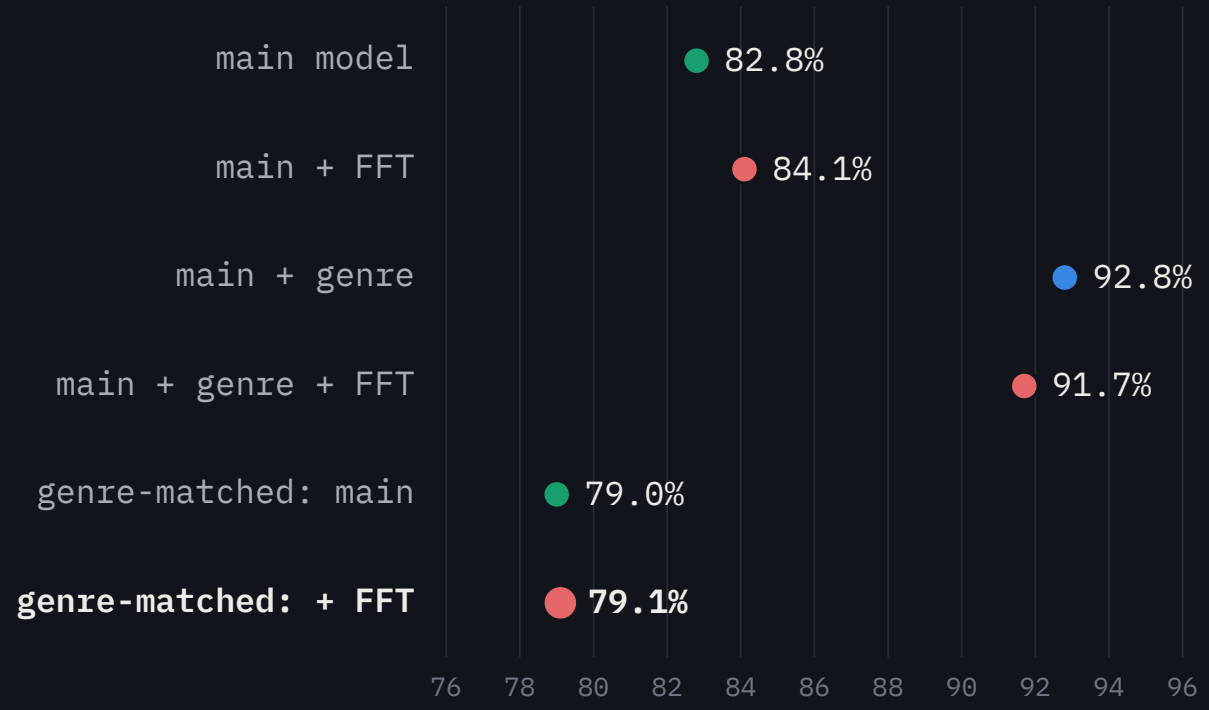
ARTEFACT **Song titles +6.7**, more than any musical feature. Titles don't make hits — they reveal that non-hits come from niche genres and catalogue releases.

LUCK **Rich librosa audio**: +3.8 (p = .002) on one split, **+0.9, not significant** in CV. Same for "PCA 40 is the peak" and the family knock-outs.

chorus_hit + sections, often suspected of leakage, add nothing. FFT survives here — the next slide shows why it should not.

AUDIT · SPECTRAL FEATURES (FFT) ON THE WHOLE BENCHMARK

FFT looked like a real gain for new artists — it was the genre



20,610 songs with a clip and a Spotify genre · 5×5 CV · paper Table X

On 35,370 songs, FFT adds **+1.4** (25/25 folds, $p = 4{\cdot}10^{-9}$), all of it on new artists.

But Latin, classical, jazz, world and metal artists make **50 % of the non-hits and 1.9 % of the hits** — by construction. Audio can recognise genre (FFT: 49.9 % vs 26.1 % by chance).

With genre known, FFT **lowers** accuracy (−1.1). On a **genre-matched sample** (as many hits as non-hits in every genre): **+0.1, p = .84**. REMOVE

Genre alone lifts accuracy to 92.8 %, but Spotify assigns genres afterwards, so we use it only as a control.

Train on one decade, test k decades later: artist history ages well, FFT does not

K	PAIRS	AUDIO	+ ARTIST	+ FFT	+ ARTIST + FFT
0 (same decade)	6	.809	.837	.831	**.859**
1 decade later	5	.765	.796	.782	**.810**
2	4	.728	**.775**	.736	**.775**
3	3	.689	**.740**	.673	.728
4	2	.666	**.734**	.661	.718
5	1	.664	**.754**	.599	.706

Audio-only accuracy falls from **.809** in the same decade to **.689** three decades later: what a hit sounds like changes.

The advantage of **artist history grows** with the gap (+2.8 → +5.1 points).

FFT helps only nearby and hurts from three decades on: it encodes era-specific sound and genre.

Random Forest, 35,370 songs with a clip. Cells average 6 to 1 decade pairs, so no significance test. Paper Table XI.

Three more checks

ARTIST MATCHING

99.99%

of hits are found in the chart history. The songs that do not match are non-hits by artists who never charted — not a matching error.

Stricter matching −0.5 ($p = 4{\cdot}10^{-4}$); counting featured artists +0.1 (n.s.). KEEP the original rule.

LYRICS · SPOTGENTRACK, 79,420 TRACKS

+.023 AUC

Six lexical descriptors (word count, vocabulary richness, …) on top of decade + artist: AUC .774 → .797, accuracy .808 → .814 ($p = .002$).

The only content feature that adds to artist history on data not selected by genre.
PROMISING

PRECISION · RECALL · F1 OF THE HIT CLASS

.787 → .831

F1 on the benchmark ($p = 7{\cdot}10^{-15}$): precision and recall both rise, so the gain is not a trade-off.

SpotGenTrack: audio finds 8 % of the hits; with decade + artist, **25 %** at the same precision (F1 .147 → .358).

SUMMARY · EVERY FEATURE WE TRIED

What we keep, what we removed, and why

FEATURE	RESULT	VERDICT	WHY
Artist history	+2.9 alone · +2.6 on ISMIR, years matched	KEEP	Survives every control
Era (decade / year)	+2.0 · useless once years are matched	KEEP	Helps new artists; on ISMIR only a date shortcut
Song title (TF-IDF)	+6.7, $p = 4{\cdot}10^{-20}$	REMOVE	Artefact of how non-hits were collected
Title length · chorus_hit + sections	+0.2 · 0.0, not significant	REMOVE	No signal
Rich librosa audio (185 → PCA 40)	+3.8 on one split → +0.9, n.s.	REMOVE	The single-split gain was luck
FFT spectral (31)	+1.4 → +0.1 genre-matched	REMOVE	Genre shortcut
Spotify artist genre	+10.1	CONTROL ONLY	Assigned afterwards: would leak
Low-level audio (207)	≈ 0 on top of artist history	REMOVE	Nothing beyond artist history
Lyrics, lexical (6)	AUC +.023, accuracy +0.6 ($p = .002$)	PROMISING	Small; indicative only
Market trend (vs last decade's hits)	+0.1, n.s.	REMOVE	Only carries the era (Part 6)

STATISTICS · ABOUT 60 P-VALUES IN ONE STUDY

With many tests, some pass by chance — so we apply Bonferroni: $p < 0.05 / 60 \approx 8 \cdot 10^{-4}$

SURVIVES — OUR CONCLUSIONS REST ON THESE

- Main gains on the benchmark ($p = 7 \cdot 10^{-17}$)
- All five classifiers
- Head-to-head with Du, all four models
- Temporal hold-out on unseen decades
- Artist history on ISMIR, protocols A and B
- Artist-grouped gains
- The genre control ($p \leq 5 \cdot 10^{-7}$)

INDICATIVE ONLY — $8 \cdot 10^{-4} < p < 0.05$

- Decade and FFT gains on the small audio subset
- Most single-split gains on that subset
- SpotGenTrack accuracy, artist alone ($p = 10^{-3}$)
- ISMIR cold-start gains ($p = .002, .003$)
- Lyrics on SpotGenTrack ($p = .002$)

Eight claims did not survive re-checking

EARLIER SLIDE SAID	WHAT IS ACTUALLY TRUE
Reference paper at 0.877	Middlebrook & Sheik — a **different dataset**, not comparable
Rich audio +3.8, “PCA 40 is the peak”	**+0.9, not significant** in CV; all PCA sizes within noise
Billboard 2000–23 validation: no gain	**Invalid**: 2,911 of 3,397 rows had no audio features
SpotGenTrack “our 2 features alone” .8222	That score was **artist + low-level audio**
Tilburg .878 / .890, “a close comparison”	Exact protocol, 50 splits: **.867 / .861** vs published .84 / .82
FFT +1.4 for new artists (our own R7)	A **genre shortcut**: +0.1 on a genre-matched sample
“Artist names match for only 52 %”	**99.99 % of hits** match; the rest are non-hits by artists who never charted
Artist history “works on new songs too”	It does **not help new artists** on the benchmark (+0.4, n.s.)

Trend, seasonality, residual — and which forecasting algorithms work

COMPONENTS · 68 YEARS OF WEEKLY CHARTS

Debuts per week: mostly **trend** (strength .72): 11 in 1959, 5.5 in 2001, 11 again with streaming. Seasonality weak (.11): January is quiet. Largest residual spike: a **rule change** in December 1998.

Holiday songs on the chart: almost pure seasonality. The **FFT** finds the 52-week cycle and its harmonics (26, 17.4, 13 weeks) without being told; a Fourier fit follows the streaming-era peaks (R^2 = .94).

Notebook `timeseries/time_series_hot100.ipynb`: STL decomposition, ACF/PACF, ADF/KPSS, Fourier terms.

FORECASTING DEBUTS 12 MONTHS AHEAD · 11 ORIGINS · MASE (LOWER IS BETTER)

ARIMA(1,1,2)	1.82	KEEP
ARIMA + Fourier terms	1.82	KEEP
Holt-Winters (damped)	1.85	KEEP
SARIMA (seasonal)	1.89	REMOVE
Gradient boosting / random forest	1.94 / 1.99	REMOVE
Seasonal naive	2.32	BASELINE

The three best are indistinguishable (p = .97) and beat the seasonal naive forecast at 10 of 11 origins (p = .006). SARIMA: the series has almost no season to model. Trees: 800 points are too few, and they cannot extrapolate a trend.

TIME SERIES FOR HIT PREDICTION

After release, the first weeks predict a top-10 hit; before release, trends add nothing

AFTER RELEASE · FIRST WEEKS ON THE HOT 100 → TOP 10?

INPUT	AUC	F1
year, month, artist history only	.59	.15
first week (debut position)	.75	.40
first 2 weeks	.85	.53
first 3 weeks	**.89**	**.61**
first 4 weeks	.92	.67

31,925 songs, 16 % reach the top 10. Random Forest on positions, best rank, slope and acceleration + context; 5×3 CV, every gain 15/15 folds, $p < 10^{-6}$.

Not trivial: on songs **not yet in the top 10** after 3 weeks, AUC **.87**; trained before 2000 and tested after, **.85**. KEEP

BEFORE RELEASE · MARKET TREND ON THE KAGGLE BENCHMARK

Distance of a song's audio to the **previous decade's hits**, and whether it moves in the direction of the trend.

+0.1 pt

on top of the main model, p = .66. REMOVE

Per-feature deviation: +0.5 (p = .006, indicative). On audio alone the trend adds +1.8 — less than the decade itself (+2.0): it only carries **era**, which the model already has.

Time series help once a song is out: its first weeks are the best predictor we found. Before release, they add nothing to artist history and era.

WHAT IS NEW

Artist history is not a new idea. Proving what it is worth is.

ALREADY EXISTED

Interiano et al. (2018): a “superstar” indicator (did the artist chart recently?) improves a Random Forest on 500,000 UK songs.

Middlebrook & Sheik (2019) list an artist past-performance field, but never isolate or test it.

Neither tests significance, compares on the same data, or checks for sampling artefacts.

OUR CONTRIBUTION

- A **leakage-free** version (strictly earlier periods)
- A significance-tested **head-to-head** on the same data as prior work
- **External validation** with a year-matched control
- **Unseen-artist** and **cold-start** analysis
- An **audit** of artefacts and single-split luck

What this study cannot say

LIMITATIONS

- Artists matched by name, not by a canonical ID (homonyms)
- No deep-learning model: we compare inputs at equal model
- Artist history partly measures **cumulative advantage**, not a song's appeal
- ISMIR: a Random Forest and a different draw of negatives, not their network
- Du: re-drawn splits, since the original R seed cannot be replayed
- Audio clips come from Deezer, not from Spotify

NEXT

- Features that help **first-time artists**: label, collaborators, early streams — always with a genre-matched control
- Semantic lyrics and learned audio embeddings, under the same audit
- Early-trajectory forecasting (Part 6) with streaming data
- A benchmark whose negatives ignore artist, genre and date

TAKE-HOME

Knowing which gains are real matters more than the headline number

+5.0

points on the Kaggle benchmark, every classifier, every decade — an upper bound inflated by how non-hits were chosen.

+2.6

points on independent data, years matched, artists unseen: the value we defend.

76.3%

above the best published 75.04 % on ISMIR 2019, and ahead of the only prior model on the benchmark on 50/50 splits.

Song titles, release dates, FFT and rich audio on one split looked better still — and were artefacts, genre or chance. Every number: `repro/ scripts → paper_results/`